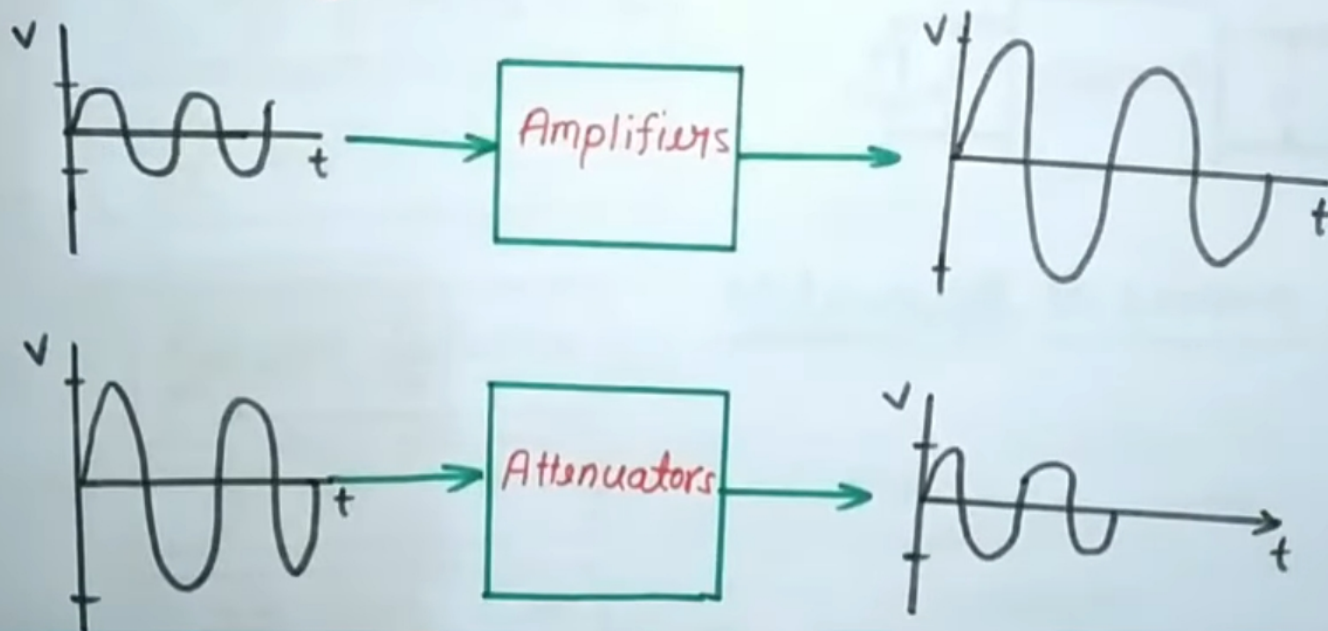


✧ ATTENUATORS ✧

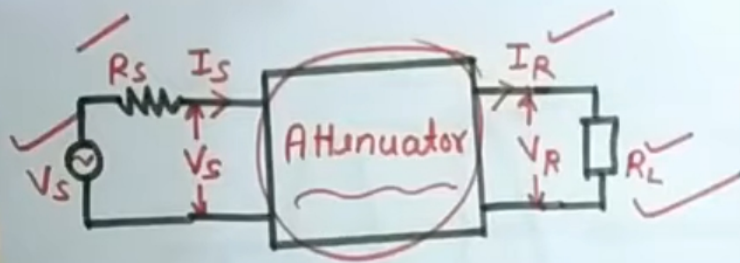
✓ "An attenuator is a TPN connected between Source and load to provide desired attenuation of the signal."



Like Comment And Share

Units of Attenuators :-

→ dB में Attenuation



$$\underline{D} = 10 \log_{10} \left| \frac{P_s}{P_R} \right| \checkmark$$

$$\underline{D} = 20 \log_{10} \frac{V_s}{V_R}$$

$$\underline{D} = 20 \log_{10} \frac{I_s}{I_R}$$

→ Nepers में Attenuation

$$\underline{\alpha} = \frac{1}{2} \log_e \left| \frac{P_s}{P_R} \right|$$

$$\underline{\alpha} = \log_e \frac{V_s}{V_R}$$

$$\underline{\alpha} = \log_e \frac{I_s}{I_R}$$

Relation b/w dB and Nepers :-

$D \rightarrow$ Attenuation in dB

$\alpha \rightarrow$ Attenuation in Nepers

$\rightarrow D = 8.686 \times \alpha$

$\rightarrow \alpha = 0.115 \times D$

Classification of Attenuators :-

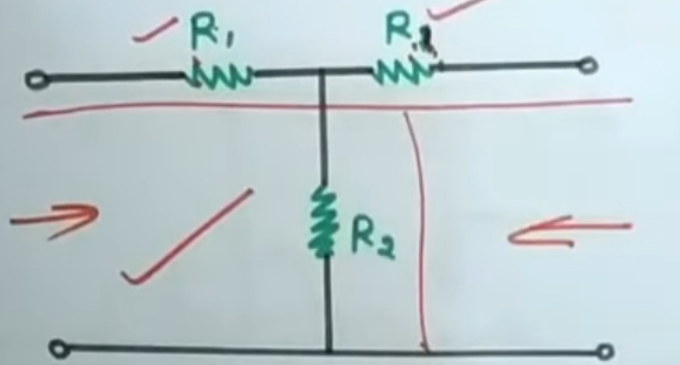
Symmetrical
Attenuators

Asymmetrical
Attenuators.

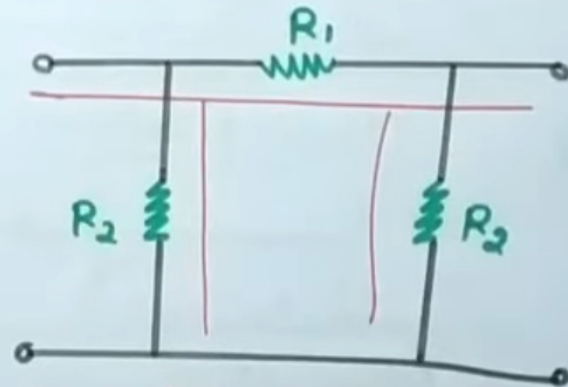
Symmetrical Attenuators :-

- यह एक Resistive N/w होता है। इनको source तथा load के मध्य Connect करते हैं।
- इन Attenuators के i/p व o/p Resistance equal होते हैं।

$$R_i = R_o$$



Symmetrical T-Attenuators



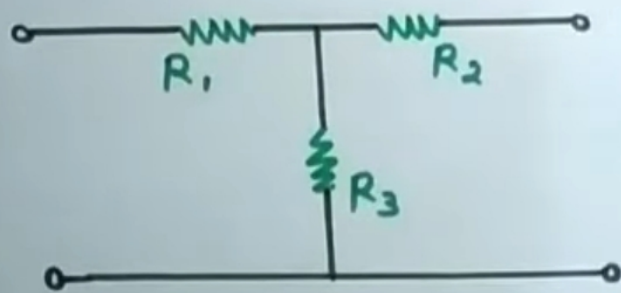
Symmetrical Pi-Attenuator

Asymmetrical Attenuators :-

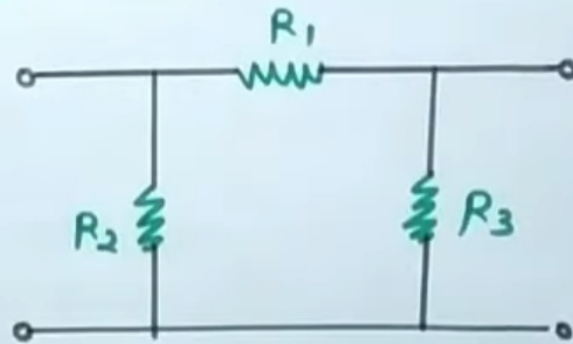
→ यह भी एक Resistive N/w होता है। इसको भी source व Load के मध्य connect किया जाता है।

→ इनमें input व output Resistance की Value different होती है।

$$R_i \neq R_o$$



Asymmetrical T-
Attenuator

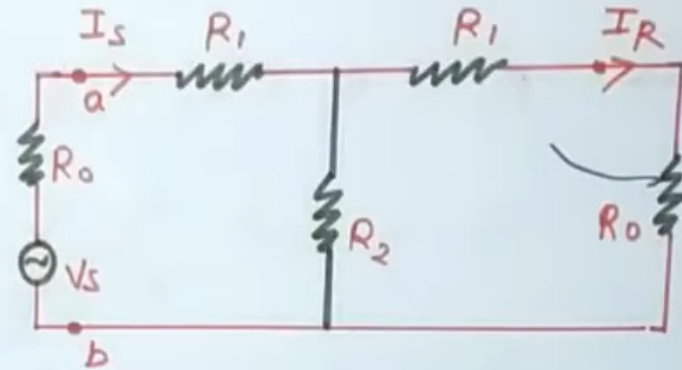
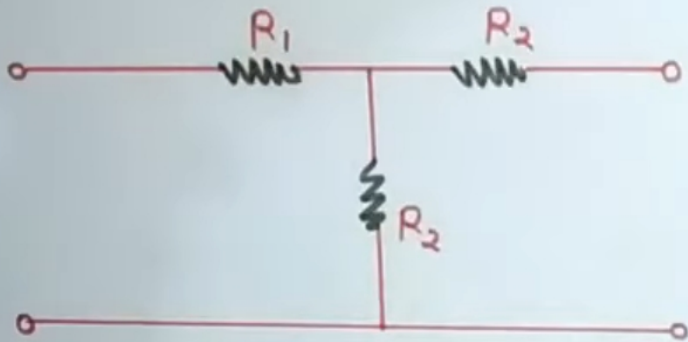


Asymmetrical
Pi-Attenuator

Application of Attenuators :- (Use)

- For Attenuation
- For Impedance Matching
- As Variable Attenuators
- For increasing input impedance of equipments.

Design of Symmetrical T-Attenuator



$$R_1 \text{ or } R_2 = ?$$

→ Current division Rule —

$$\Rightarrow I_R = \frac{R_2}{R_1 + R_2 + R_0} \cdot I_S, \quad \left[N = \frac{I_S}{I_R} \right]$$

$$\Rightarrow N = \frac{R_1 + R_2 + R_0}{R_2} = 1 + \frac{R_1}{R_2} + \frac{R_0}{R_2}$$

$$\underline{N-1} = \frac{R_0 + R_1}{R_2} \quad \text{eqn ①}$$

$$\rightarrow \underline{R_0} = \underline{R_1} + [\underline{R_2} \parallel (\underline{R_1} + \underline{R_0})]$$

$$\underline{R_0} = \underline{R_1} + \frac{R_2(R_1 + R_0)}{R_1 + R_2 + R_0}$$

$$\underline{R_0} = \underline{R_1} + \frac{R_2 + R_0}{\frac{R_1 + R_2 + R_0}{-R_2}}$$

$$R_0 = R_1 + \frac{R_1 + R_0}{N}$$

$$NR_0 = NR_1 + R_1 + R_0$$

$$NR_0 - R_0 = NR_1 + R_1$$

$$R_0(N-1) = R_1(N+1)$$

$$R_1 = \frac{R_0(N-1)}{N+1}$$

R_1 की Value eqⁿ ① में put करने पर

$$\Rightarrow N-1 = \frac{R_0 + R_0 \left(\frac{N-1}{N+1} \right)}{R_2}$$

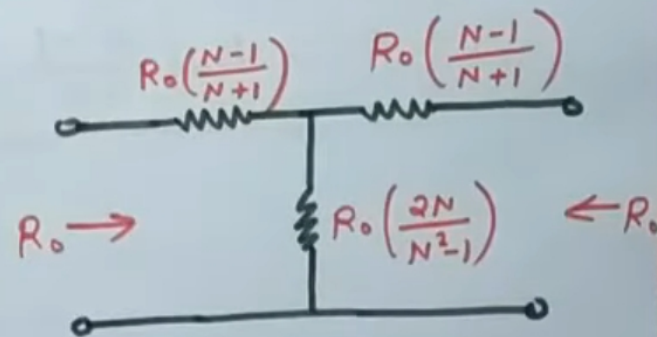
$$\Rightarrow R_2 (N-1) = \frac{R_0 (N-1) + R_0 (N-1)}{(N+1)}$$

$$\Rightarrow R_2 (N-1) = \frac{2R_0 N}{N+1}$$

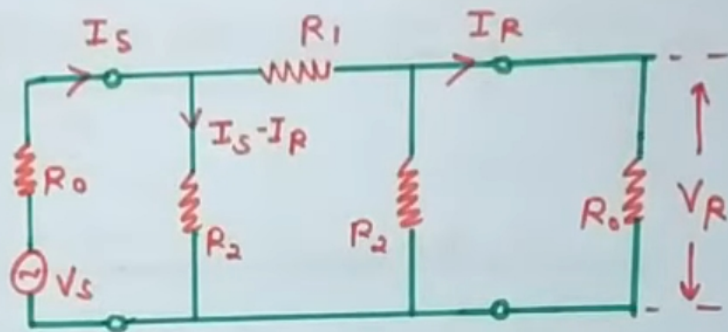
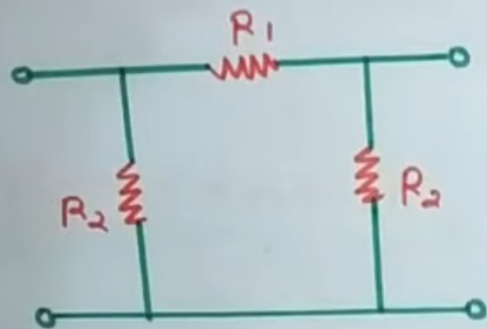
$$\Rightarrow R_2 = \frac{2R_0 N}{(N+1)(N-1)}$$

$$\Rightarrow \boxed{R_2 = R_0 \frac{2N}{N^2-1}}$$

Design Eqⁿ



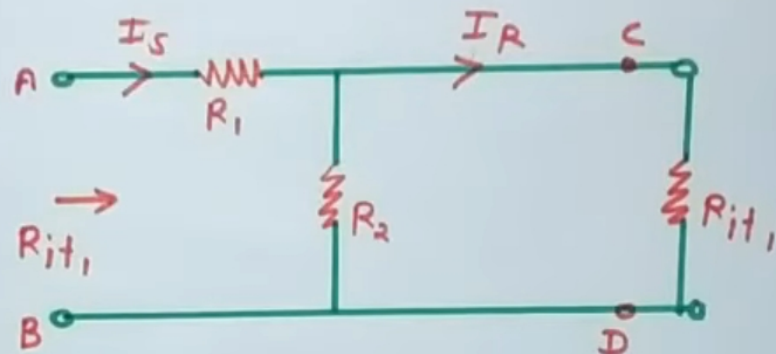
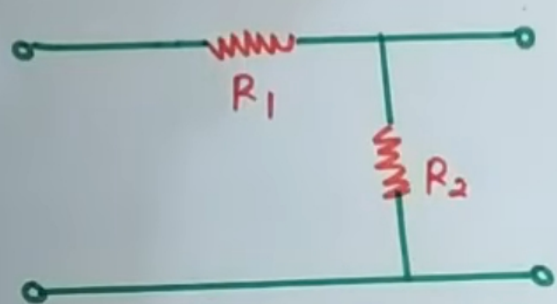
Design of Symmetrical π -Attenuator :-



$$R_1 = R_0 \left(\frac{N^2 - 1}{2N} \right)$$

$$R_2 = \left(\frac{N + 1}{N - 1} \right)$$

L-type Attenuator : - (Designing)



$$R_1 = \left(\frac{N-1}{N} \right) R_{it,1}$$

$$R_2 = \frac{R_{it,1}}{N-1}$$